Kamalnath K

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Objective

Computer Science graduate with a keen interest in applying foundational knowledge to real-world challenges. I seek an environment that fosters learning and allows me to contribute to the growth of the organization and myself.

Education

B.E. Computer Science and Engineering ***Oct 2022 – Apr 2026***
Anna University(CEG), Chennai ***CGPA: 7.53***
Higher Secondary Education ***Jun 2021 – Jun 2022***
Gokilambal Hr Sec School,Ariyalur ***Percentage: 97.1%***
Secondary Education ***Jun 2019 – Jun 2020***
Gokilambal Hr Sec School,Ariyalur ***Percentage: 93.4%***

Skills

Programming Languages: Python,Java
Frameworks & Technologies: ReactJS, ExpressJS,Flask
Cloud Technologies: AWS(EC2, ELB, IAM),GCP(Basics)
DevOps&Tools: Git,SSH (PuTTY),Terraform
Databases: PostgreSQL,MongoDB
Soft Skills: Teamwork, Time Management,Adaptability,Crowd Management

Projects

Continuous Sign Language Recognition and Cross-Sign Conversion ***Apr 2026***

- Developed a system to recognize continuous Indian Sign Language (ISL) and convert it into text for better communication with non-signers using BilSTM amd converts ISL gestures into Other Sign Language for broader accessibility.
- Implemented Huffman-based segmentation, MediaPipe keypoint extraction, Bi-LSTM sequence modeling, and 3D avatar-based cross-sign translation.
- Integrated Transformer-based sentence generation and 3D avatar-based cross-sign translation.
- **Tech Stack:** Python, MediaPipe, Three.js, Flask,

Osteoporosis Prediction System ***Sep 2025***

- Developed a machine learning system to predict osteoporosis risk from clinical and ultrasound-based features.
- Applied data preprocessing, feature selection, and ensemble machine learning techniques to improve prediction performance.
- Achieved 97% binary prediction accuracy and 92% multi-class classification accuracy for osteoporosis risk assessment.
- **Tech Stack:** Python,Streamlit,pandas

Order and Inventory Management System ***Apr 2024***

- E-commerce web app that manages inventories and streamlines ordering and billing.
- Designed a relational database system to manage products, customers, orders, inventory, and billing for a retail shop.Enabled real-time purchase, cart management, and order processing, reducing manual effort by 70
- **Tech Stack:** ReactJS, ExpressJs, PostgreSQL

Positions Of Responsibilities

Students Association and Arts Society(SAAS)- Organiser ***Mar 2025 – Present***

- Contributed to organizing CEG's 24' Cultural Event in collaboration with Zebronics and Krafton.

CEG Tech Forum(CTF)-Human Resources (HR) Coordinator ***Sep 2024 – Apr 2025***

- Contributed to the development of the **Kurukshetra tech fest** Assisted in volunteer onboarding and participant

Certifications

Data Analytics by Google - TN Upskilling(InGage)
AWS Cloud Foundations - Amazon Academy